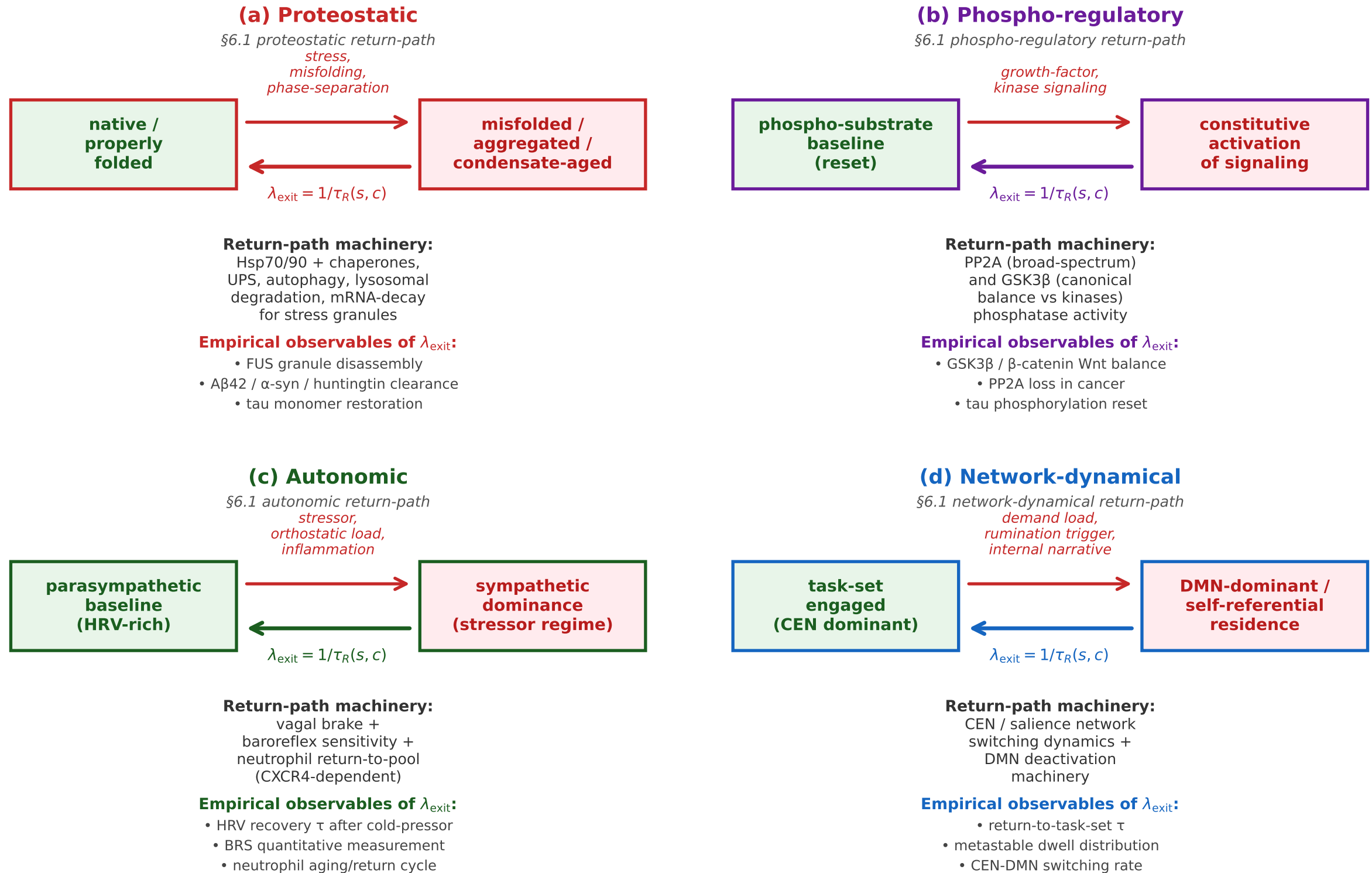


Figure 3. Return-path machinery typology — four substrate-specific instances of $\lambda_{\text{exit}} = 1/\tau_R(s, c)$



All four classes share the same observable structure: a functional baseline state, a perturbed state, and a return-rate-constant λ_{exit} set by substrate-specific machinery (§6.1). Disease at any substrate corresponds to fall of λ_{exit} below the threshold needed to maintain $R(s, c) = \tau_R(s) \times \pi_c$ within functional bounds.